CLAIMS

What is claimed is:

1. A method for providing forward secrecy in a communication session between two devices using disposable encoding maps, the method comprising the steps of:
    - commencing data transmission for the session using a first data encoding map;
    - monitoring for the occurrence of a pre-defined trigger condition;
    - upon occurrence of the trigger condition, ceasing to use the first data encoding map and commencing use of a second, different data encoding map for subsequent data transmission in the same session; and
    - discarding the first data encoding map.
2. The method of claim 1, wherein the trigger condition is the transmission of a pre-defined number of data frames.
3. The method of claim 1, wherein the trigger condition is the expiration of a pre-defined time interval.
4. The method of claim 1, wherein the first and second data encoding maps are generated independently by each device based on a shared secret and a deterministic algorithm.
5. The method of claim 4, wherein the second map is the next map in a pseudo-randomly generated sequence of maps.
6. The method of claim 1, wherein the communication session is conducted over a parallel communication channel with a cardinality greater than 2.
7. The method of claim 6, wherein the parallel communication channel is a massively parallel optical interface as described in U.S. Patent Application No. 19/559,705.